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# Therapeutic Exercises and Modalities in Athletes With Acute Hamstring Injuries: A Systematic Review and Meta-Analysis

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**Context:** Hamstring strain is a common injury to the lower limbs. Early intervention in the acute phase aids with restoring hamstring function and prevents secondary related injury.

**Objective:** To systematically review and summarize the effectiveness of exercise-based interventions combined with physical modalities currently used in athletes with acute hamstring injuries.

**Data Sources:** Five databases (EMBASE, Medline, Cochrane Library, SPORTDiscus, and Web of Science) were searched from inception to July 2021.

**Study Selection:** A total of 4569 studies were screened. Nine randomized controlled trials (RCTs) on the effect of therapeutic exercise programs with and without physical agents in athletes with acute hamstring injuries were identified for meta-analysis.

**Study Design:** Systematic review and meta-analysis

**Level of Evidence:** Level 1

**Data Extraction:** The studies were screened, and the evidence was rated using the PEDro scale. Nine RCTs with PEDro scores ranging between 3 and 9 were included and extracted pain intensity, time to return to play (TTRTP), and reinjury rate in the study.

**Results:** Loading exercises during extensive lengthening were shown to facilitate TTRTP at $P < 0.0001$ but did not prevent recurrence ($P = 0.17$), whereas strengthening with trunk stabilization and agility exercise did not reduce the duration of injury recurrence ($P = 0.16$), but significantly reduced the reinjury rate ($P < 0.007$) at a 12-month follow-up.
The results of the stretching programs and solely physical modalities could not be pooled in the statistical analysis.

**Conclusion:** The meta-analysis indicated that a loading program helps athletes to return to sports on a timely basis. Although strengthening with trunk stabilization and agility exercise cannot significantly reduce recovery time, the program can prevent reinjury. The clinical effects of stretching programs and pure physical modality interventions could not be concluded in this study due to limited evidence.

**PROSPERO Registration:** CRD42020183035

**Keywords:** acute hamstring injuries; pain; physical modalities; therapeutic exercise; time to return to play; reinjury rate

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Hamstring strain injuries are among the most common sports injuries in both contact and noncontact sports, with rates as high as 37% compared with other lower extremity injuries.[16] The prevalence of such injuries is increasing by 4% per year, and the injury rate during training tends to increase to a greater degree than is the case during competitive sport.[15] An estimated 3 to 4.1 injuries per 1000 hours of competition and 0.4 to 0.5 injuries per 1000 hours of training have been reported among both amateur and professional players.[32] This type of injury has a high recurrence rate: 12% to 48% in English professional soccer and over 20% in Australian football.[23,57] This injury also leads to absences from sports events and extensive rehabilitation periods. For example, 17.2 to 18.8 absence days have been reported in professional football players,[14] and 6 to 50 weeks (median 16 weeks) in high-speed runners.[3] Returning to play along with the prevention of reinjury are 2 main targets of hamstring rehabilitation.[59] This persistent injury puts both athletes and the team management under pressure to allow players to return to competition as quickly as possible. Hence, a systematic evaluation of the current rehabilitation programs may lead to more effective treatment and management of acute hamstring injuries.

Therapeutic exercise plays an important role in a rehabilitation program for hamstring injuries. The findings of previous studies have suggested that incorporating therapeutic exercise with other therapies in the early stages of such injuries helps shorten the length of recovery after an injury and facilitates rehabilitation protocols.[52] Eccentric contraction is an integral part of lower extremity movements in sports, especially during sprinting and kicking. This type of muscle action is also a common cause of injuries to the hamstring.[30] Eccentric exercise has been proven to improve muscle strength characterized by muscle microlesions and greater mechanical tension compared with other forms of contractions,[22] which induces morphological, physiological, and neural adaptations,[17,33] and prevents damage to the hamstring.[49] Following successful rehabilitation programs, and being given return to play (RTP) clearance, chronically injured players still have been found to have deficits in fascicle length, muscle strength, reflex response, and muscle activation.[19,44] Compared with a general rehabilitation program, a hamstring rehabilitation program should focus more on the acute phase, which requires more specific exercises. Good cooperation between injured athletes and therapists is crucial to facilitate time to returning to the previous level of sports performance and eliminating factors related to risk of reinjury.

Several previous studies reported the efficacy of exercise programs for hamstring injuries.[43,55] Pas et al[43] published a systematic review and meta-analysis of conservative interventions, including exercises, platelet-rich plasma (PRP) injections, manipulation, and nonsteroidal anti-inflammatory drug (NSAIDs). It was reported that a loading program reduced the time to return to play (TTRTP) and recurrence, whereas PRP injections, manipulation, or even NSAIDs had no effect on outcomes following acute hamstring injuries. Due to the limited evidence, the efficacy of trunk stabilization exercises and agility training was not concluded in the meta-analysis. From the latest published evidence, no new literature reviews have been carried out among athletes with acute hamstring injuries.

Physical modalities are considered to be part of a standard of care in the management of acute soft tissue injuries.[46,58] The use of physical modalities at the initial phase of a sports injury shortens the inflammation period, promotes healing at the cellular level, shortens time to return to activities, and prevents secondary hypoxic injuries.[11] Takenori et al[54] reported the efficacy of low-level laser on immediately relieving pain by 28.74% and pain relief of 75% after a program carried out in college athletes with motion pain. In addition, Sefiddashti et al[50] confirmed that cryotherapy with a lengthening program led to better improvement of hamstring range of motion (ROM) and function in athletes with hamstring injuries. Despite the fact that the application of physical agents for the management of hamstring strain injuries has been suggested and broadly applied,[45] evidence supporting specific modalities among competitors with acute hamstring injuries has not yet been documented. In addition, there has been no consensus about the therapeutic effects of physical modalities on pain reduction in athletes with hamstring injuries.

A combination of physical agents and therapeutic exercise has been demonstrated to have a superior effect as compared with exercise alone. Nevertheless, evidence of the effects of therapeutic exercise combined with physical agents specifically directed toward acute hamstring injuries has not been documented in the review literature. There remains a controversy as to the role of physical agents on the effectiveness of rehabilitation programs.[38] Conclusive information on the effects of physical agents combined with therapeutic exercise may be an integral part of the decision-making process for therapists and multidisciplinary teams in sports injury clinics. Therefore, the objective of this meta-analysis is to scrutinize and update the current rehabilitation protocols including a key component of therapeutic exercise combined with physical modalities that are commonly used in physical therapy clinics compared with other forms of exercise employed to mitigate pain and reinjury rate, as well as to shorten time to return to sport in athletes with acute hamstring strain injuries.

## METHODS

### Literature Search

The search was performed based on the method in the Cochrane Handbook and reported according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines by 2 investigators.[25,42] The EMBASE, Medline, Cochrane library, SPORTDiscus, and Web of Science databases were used from inception to July 2021 (Appendix 1, available in the online version of this article). All studies were searched and imported to a citation database (EndNote X9.2), and duplicates were removed. The search terms are listed in

Table 1. Summary of keywords employed in the database search

| Muscle Group | Injury | Intervention and Modalities | Outcomes |
|---|---|---|---|
| hamstring*<br>"hamstring muscle*"<br>"posterior thigh"<br>"Biceps Femoris"<br>Semimembranosus<br>Semitendinosus | injur*<br>strain*<br>rupture<br>tear | therapy<br>exercise<br>"therapeutic exercise"<br>intervention<br>"therapeutic intervention"<br>training<br>"training intervention"<br>"physical modalit*"<br>"physical agent*"<br>"physical instrument*"<br>"physical apparatus*" | Pain<br>"return to play"<br>"return to sport*"<br>"re-injur*" |

Table 1. The study was registered to the international prospective register of systematic reviews. The PROSPERO registration number was CRD42020183035.

## Inclusion Criteria

Studies were selected following the criteria defined based on the PICO search tool.[39] The inclusion criteria of the study included athletes specializing in all types of sports with acute hamstring injuries where they were treated using therapeutic exercise programs, physical modalities, or a combination of both forms of treatment. Essential key exercise programs including isometric, concentric, and eccentric exercise were compared with no program, standard treatment modalities, or different programs. Treatment with physical modalities included at least 1 type of physical modality compared with no treatment, standard treatment, or different modalities. The studies that were included had to provide at least 1 of the following outcome measures (pain scores, TTRTP, or reinjury rate). These outcomes were compared with the results for a control group and included group effects and/or pre- and postintervention. Included were only randomized controlled trials (RCTs) in English.

## Exclusion Criteria

Studies were excluded if they were systematic reviews, meta-analyses, cross-sectional studies, conference abstracts, case reports, case series, observational studies, or were not RCTs. The participants were required to have been diagnosed with a hamstring avulsion injury or chronic injuries. Studies that did not report any training or intervention were also excluded. It is important to note that studies were not excluded based on the characteristics of the participants, such as age, gender, training hours, and sports skill.

## Quality Assessment

The PEDro scale was used to assess the quality of the included studies. The scale identifies internal validity, external validity, and sufficient statistical information to make the results interpretable. The scale consists of 11 questions with 10 scores and includes eligibility criteria, random allocation, concealed allocation, baseline comparability, blind subjects, blind therapists, blind assessors, adequate follow-up, intention-to-treat analysis, between-group comparisons, and point estimates and variability. Note that the eligibility criteria item (item 1) did not contribute to the total score. The scores could be interpreted according to 4 categories, where excellent quality was identified as a score of 9 to 10; good quality was identified as a score of 6 to 8; fair quality was identified as a score of 4 to 5; and poor quality was identified as a score of 3 or less. The quality assessment was extracted from the PEDro database (from the PEDro website, https://pedro.org.au/english/resources/pedro-scale/). All the scored papers in the database had been confirmed by PEDro committees. In cases where the included studies were not assessed by the database, the papers were scored by 2 reviewers independently following PEDro quality assessment guidelines (a list of criteria for quality assessment of randomized clinical trials for conducting systematic reviews based on a Delphi consensus).[36] When the scores were not agreed upon by the 2 reviewers, a third reviewer was consulted to resolve conflicts.

## Data Analysis

Data from the 9 included studies were extracted and entered independently into a Microsoft Excel 2019 spreadsheet (Microsoft Corporation) by 2 reviewers. Descriptive and categorical data were also extracted and entered into a Microsoft Excel spreadsheet for comparison and interpretation of the findings. The extracted data included the sample sizes in each allocation group, participant characteristics (age, sex, and sports types), diagnoses of the participants, the type of assessments conducted, the frequency of the intervention program, and the follow-up period. All the extracted data were in agreement from the 2 authors for further analysis.

The results for the pain score were based on the severity of pain at the injured area using the Visual Analog Scale (VAS) or

Numerical Rating Scale (NRS)–a psychometric measurement instrument that measures a characteristic or attitude of patients believed to range across a continuum of values, with 11-point Likert scales with scores ranging from 0 to 10, where a minimum score meant no pain and a maximum score meant the worst pain imaginable.[29] TTRTP refers to the number of days beginning from the first training day to the return to practice or competition based on the decision-making process (injured athletes are declared safe to return to a training program or a competition) after a health examination carried out by doctors, therapists, or coaches.[56] The reinjury rate was defined as the number of repeated episodes of injury after treatment programs in subsequent follow-up periods.[18]

Continuous data (means and standard deviations) and *P* values for pain score and TTRTP were extracted and then synthesized and analyzed using Review Manager (RevMan) version 5.3 (The Nordic Cochrane Centre). The weighted mean difference (WMD) was a pooled estimate and presented in the form of a forest plot. The WMD was calculated from the final data result (intervention group) after subtracting the initial data (relative control group).[25]

Risk difference (RD) was used to assess the reinjury rate between the intervention groups and the relative control groups. The value was set at 0.95 for pain and TTRTP and 0.1 for the reinjury rate with respective 95% CI.[48] The calculation method was performed based on the following formula: $RD = IR_e - IR_u$, where $IR_e$ is the reinjury rate among the intervention groups and $IR_u$ is the incidence rate among the relative control groups).[31] The result of RD can be interpreted as either positive (increased risk) or negative (decreased risk by the exposure). For example, an RD of -0.15 highlights subjects who attended an exercise program who had 1.5 fewer cases of reinjury per 100 people compared with subjects who did not participate in an exercise program or a control group.

Heterogeneity was considered based on the inconsistency index ($I^2$) expressed as a percentage, which was computed from $I^2 = 100 \times [(Q - df)/Q]$, where $Q$ is Cochran's heterogeneity statistic and $df$ is the degrees of freedom.[27] The value of $I^2$ is from 0% to 100% and can be interpreted as follows: 0% to 40%, might not be important; 30% to 60%, may represent moderate heterogeneity; 50% to 90%, may represent substantial heterogeneity; and 75% to 100%, represents significant heterogeneity.[25] If the value of $I^2$ is 0% to 50%, a fixed effects model will be used, but if the value of $I^2$ is more than 50%, the random effects model will be used to estimate the outcome parameters.[26]

# RESULTS

## Literature Search

The search strategy was conducted following PICO guidelines.[39] Figure 1 highlights the flow diagram for searching, which consisted of 5 steps: A total of 5751 articles were retrieved from 5 databases (Embase 2249; Medline 1625; Cochrane 1071; SPORTDiscus 440; Web of Science 366) and 2 articles were added after a manual search. After the removal of duplicate articles, all 4569 articles were screened independently based on title and abstract by 2 investigators, and 24 articles that matched the inclusion criteria requiring an assessment of full-text articles were included. We also examined each complete article, and a further 15 articles were excluded for reasons presented in the flowchart. The final 9 articles were included in the quantitative synthesis.

## Quality Assessment

Six studies were extracted from the PEDro database.[5,6,36,37,50,51,53] All scores in the 6 studies extracted from the website had been confirmed by PEDro's committees. The other 3 studies were scored by 2 independent reviewers,[20,24,38] and the third reviewer followed the PEDro quality assessment guidelines. The minimum score in this study was 3, and the maximum score was 9, with a median score of 4 (Table 2). There was only 1 excellent study (total score of 9) while 2 studies were considered to be of good quality. Five studies were considered to be fair quality, with the score ranging from 4 to 5, which accounted for most of this study, and another was considered to be of poor quality (a score of 3).

## Participants

A total of 396 participants who were diagnosed with acute hamstring injuries were included, with a mean 44.00 ± 21.84 participants per study (minimum 22, maximum 80 per study). There were 317 men in this study, accounting for 93.69% of participants, and 79 women (6.31%). Only 2 studies reported the proportion of men and women in each group.[50,51] The average minimum age of the participants was 19.0 ± 3.0 years, and the average maximum age was 30.36 ± 7.06 years. The average age of the participants in this study was 24.63 ± 2.98 years. Two articles were examined according to specific type of sport (football players, sprinters, and jumpers),[5,6] while 7 studies included all sports types in the experiments. Five studies reported the severity of the injuries as grades 1 to 2,[20,37,38,50,51] and 4 studies reported only acute hamstring injuries with no specific injury classification.[5,6,24,53] Most studies assessed the participants through a physical examination. In addition, 3 studies evaluated MRI results to assess participants' injuries,[5,6,53] and 2 studies confirmed the injuries through ultrasound.[20,37] The definition of acute hamstring injury varied significantly across the studies, starting from 2 to 10 days (2, 5, 7, and 10 days in 4,[5,6,37,50] 1,[38] 1,[24] and 2 studies,[51,53] respectively). Hagag et al[20] also did not provide a clear definition of acute injury. The majority of the studies (4 studies) reported the follow-up period for the reinjury outcome to be 1-year after the TTRTP,[5,6,51,53] while the other 2 studies followed the treatment effects 6 months after the TTRTP.[24,38]

## Interventions and Outcomes

The majority of the included studies evaluated the efficacy of therapeutic exercise, including strengthening and stretching exercises. One study assessed the role of a stretching exercise only.[37] Three studies principally assessed the efficacy of

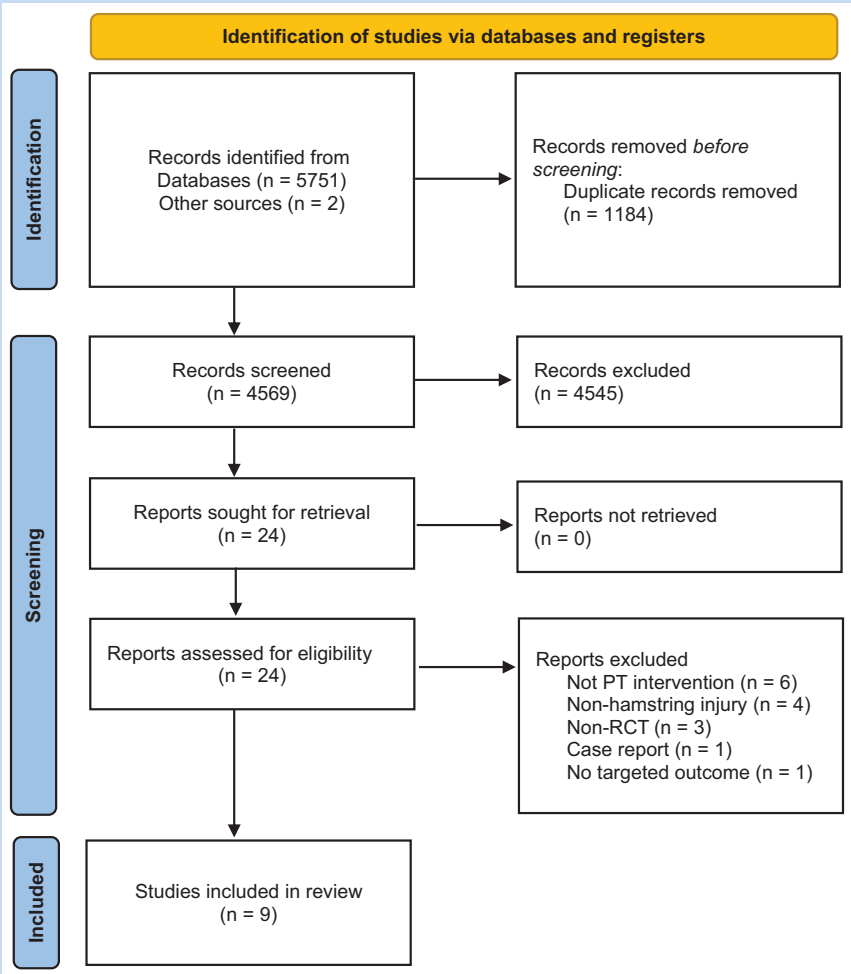


Figure 1. Flow diagram of articles search and study selection. PT, physical therapy; RCT, randomized controlled trial.

hamstring exercise combined with trunk and agility exercise.[24,51,53] Two papers examined a hamstring loading program during a lengthening exercise.[5,6] From the literature search, however, only 3 studies evaluated the physical modalities and included low-level laser therapy (LLLT) and cryotherapy combined with exercise interventions.[20,38,50] The duration of the training programs varied because all of the included studies used the TTRTP as a primary outcome measure, so the training period depended on individual compliance. Exercise compliance, which was used to determine the treatment implementation, was tracked independently by activity logbooks or telephone calls during launch days. It was interesting to note that the exercise programs were performed without pain in all studies except for 1 group in Hickey's study,[24] where the amount of pain was 1 factor in 1 group in the study (pain-threshold group).

## Data Synthesis

The main purpose of this study was to focus on the results of physical therapy interventions, including therapeutic exercises and physical modalities that are commonly used for sports injuries in rehabilitation clinics. The programs reviewed in this study can be divided into 4 main kinds of physical therapy interventions as follows: (1) lengthening or stretching hamstring exercise, (2) isolated eccentric exercise or loading exercise during extensive lengthening aimed to apply a load to lengthen the hamstring muscle-tendon units, (3) an integrated hamstring strengthening program with trunk stabilization and movement agility aimed at improving the hamstring and its proximal control of the lumbopelvic muscles, and (4) purely physical modality interventions that measured the effectiveness of the modalities at the hamstring muscle.

Based on the program reviews, the authors found that all included studies compared 2 or more different programs or different treatments, eg, combined with physical agents, which means that there were no real control groups in the included studies. However, in this meta-analysis, the authors focused mainly on the most effective programs, so all data were pooled between the studies with the different control programs. The term "relative control group" was defined as other programs or

Table 2. Summaries of the included studies rated by the PEDro scale

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | Total Score | Level of Evidence |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Malliaropoulos et al[37] | Yes | Yes | No | No | No | No | No | No | No | Yes | Yes | 3 | Poor |
| Sherry and Best[51] | Yes | Yes | No | No | No | No | No | Yes | No | Yes | Yes | 4 | Fair |
| Silder et al[53] | Yes | Yes | No | No | No | No | No | No | Yes | Yes | Yes | 4 | Fair |
| Askling et al[6] | Yes | Yes | No | Yes | No | No | No | Yes | No | Yes | Yes | 5 | Fair |
| Askling et al[5] | Yes | Yes | No | No | No | No | No | Yes | No | Yes | Yes | 4 | Fair |
| Hagag et al[20] | Yes | Yes | No | Yes | No | No | No | No | No | Yes | Yes | 4 | Fair |
| Sefiddashti et al[50] | Yes | Yes | Yes | Yes | No | No | No | Yes | Yes | Yes | Yes | 7 | Good |
| Hickey et al[24] | Yes | Yes | Yes | Yes | Yes | No | Yes | Yes | Yes | Yes | Yes | 9 | Excellent |
| Medeiros et al[38] | Yes | Yes | Yes | Yes | Yes | No | Yes | Yes | No | Yes | Yes | 8 | Good |

interventions that the studies compared, with the key programs called "intervention groups." All program details are summarized in Table 3.

## Programs Reviews

### Stretching Exercise With Physical Modalities

Two studies with quality assessment scores of 3 and 7 evaluated the same stretching program, where 1 study added cold therapy to the intervention. Due to the different outcome measures, we could not pool any outcomes.

Malliaropoulos et al[37] compared different amounts of static stretching exercise (sustained for 30 seconds per time, 4 times per session) in standing position with a full knee ROM and the trunk leaned forward. Group A stretched 1 session per day, and group B stretched 4 sessions per day. The time required for full rehabilitation was 15.05 ± 0.81 days in group A and 13.27 ± 0.71 days in group B ($P < 0.001$).

The 2 groups in the Sefiddashti's study were similarly given 20 minutes of cold therapy for 5 consecutive days with an additional stretching exercise after the cold treatment for 1 group (cryotherapy and cryostretching [CS] group).[50] In the combined CS treatment group, participants were instructed to do a stretching program, following a set protocol,[37] every 3 hours, totaling 4 to 5 times daily at home (1 session = 30 seconds of stretching followed by resting for 10 seconds, 3 sessions per 1 time). The gain score of pain showed a greater improvement in pain severity in the CS group than in the cryotherapy group, but no between-group differences were found for treatments at rest (0.74 ± 0.99 in the cryotherapy group and 1.27 ± 1.01 in the CS group, $P = 0.11$) or during activity (2.16 ± 1.64 in the cryotherapy group and 2.78 ± 1.48 in the CS group, $P = 0.24$).

## Loading Hamstring Exercise During Extensive Lengthening

Two studies with PEDro scores of 4 and 5 were conducted in the same rehabilitation program that focuses on the loading hamstring exercise during eccentric training in different targeted populations.

Askling et al[5,6] assessed the conventional (C-protocol) and the lengthening (L-protocol) program in Swedish elite football players and Swedish elite sprinters and jumpers every day, and followed once a week until the criteria for return to sport were met. Both rehabilitation protocols consisted of 3 proposed exercises, including increasing flexibility, a specific hamstring strengthening exercise, and strengthening combined with a trunk and pelvis stabilization exercise. The C-protocol, with less emphasis on lengthening, consisted of a contract/relax stretching, cable-pendulum, and pelvic tilt exercise. The L-protocol focused specifically on hamstring loading during extensive lengthening (eccentric contraction), such as the extender, the diver, and the glider. The mean for time to return to sport was 51 ± 21 and 28 ± 15 days in the C- and L-protocols, respectively, in elite football players, with no reported $P$ value, and 86 ± 34 and 49 ± 26 in the C- and L-protocols, respectively, in elite sprinters and jumpers with $P < 0.01$. Reinjury was reported in 1 case out of 38 subjects in the C-protocol and in 0 cases out of 37 subjects in the L-protocol among the elite football players (no $P$ value reported), in 2 cases out of 28 subjects in the C-protocol and in 0 cases out of 28 subjects among the L-protocol in elite sprinters and jumpers (no $P$ value reported).

A summary of the L-protocol presented significantly decreased TTRTP at $P < 0.0001$ (Figure 2; mean difference = -28.29, $Z$ = 4.17, 95% CI -41.60 to -14.99) compared with the C-protocol, but it did not differ in terms of recurrence among these 2 protocols with an RD of -0.05 (Figure 3; 95% CI -0.11 to 0.02, $Z$ = 1.38, $P$ = 0.17). Statistical heterogeneity was found in the TTRTP outcome ($I^2$ = 58%).

## Hamstring Strengthening Program Combined With Trunk Stabilization and Movement Agility With or Without Modalities

Four studies (PEDro quality range, 4-9) were categorized in this domain. Two studies mainly compared the effect of different exercise programs consisting of hamstring strengthening, trunk stabilization, and movement agility programs,[51,53] whereas 1 study examined the same rehabilitation protocol but measured differences in pain while performing the protocol.[24] Another study added a laser treatment into the protocol.[38]

Sherry and Best[51] examined subjects who did the stretching and strengthening (STST) program and the progressive agility and trunk stabilization (PATS) program as a daily home program together with follow-up once a week at the rehabilitation clinic. Both programs were divided into 2 phases. Participants in the STST group performed static stretching with isolated progressive resistance exercise in phase 1 and progressed to dynamic stretching with concentric and eccentric strengthening in phase 2. Participants in the PATS group performed agility and trunk stabilization exercises focused mainly on maintaining the spine and pelvis in a neutral alignment. They performed in the frontal and transverse plane in phase 1 and progressed to the transverse and sagittal plane in phase 2. Both groups received ice treatments for 20 minutes after finishing the exercise part of the program. Means of 37.4 ± 27.6 and 22.2 ± 8.3 days in the STST and PATS group, respectively, were reported from the first day until returning to sports activities, with no statistically significant differences found ($P = 0.25$). However, after the 1-year follow-up, participants in the PAST program showed reduced reinjuries to a greater extent than participants in the STST program, with a significant difference of $P < 0.001$ (0/13 in the PAST and 1/13 in the STST).

Silder et al[53] compared a progressive running and eccentric strengthening (PRES) program and a modified PATS 5 days per week at home and clinic visits once a week for monitoring and reevaluation of the exercise program. The programs were divided into 3 phases. Phase 1 of PRES, which was modeled on the study of Baquie and Reid[8], consisted of a hamstring isometric exercise and a short-stride jog. The program incorporated concentric and eccentric exercise in phase 2 and progressed to intense eccentric and power training with

Table 3. Results of the evidence synthesis from the reviews

| Study | N | allocation | Age (years) | Sex (M:FM) | Sports | Participants | Acute define (days) | Confirmed by | Relative control group | Intervention group | Follow-up (month) | Outcomes |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Malliaropoulos et al.[37] | 80 | A 40<br>B 40 | A 20.6 ± 3.7<br>B 20.3 ± 3.3 | 52/28 | all types | athletes with a second degree of hamstring strain | 2 | clinical evaluation and ultra-sonographic diagnosis | A: static stretching one session daily | B: static stretching four sessions daily | - | TTRTP,ROM equalization |
| Sherry and Best[51] | 24 | STST 11<br>PATS 13 | STST 24.3 ± 12.4<br>PATS 23.2 ± 11.1 | 18/6 | all types | athletes with acute hamstring strain grades 1 and 2 | 10 | physical examination | STST:<br>P1: static stretching, isolated progressive resistance exercise, icing<br>P2: dynamic stretching, concentric and eccentric strengthening | PATS:<br>P1: PATS exercises and icing in frontal and transverse plane<br>P2: PATS exercises and icing in transverse and sagittal plane | 12 | TTRTP, Re-injury, Functional testing |
| Silder et al.[53] | 24 | PRES 13<br>PATS16 | 24 ± 9 | 19/5 | all types | athletes with suspected HI | 10 | physical examination and MRI | PRES:<br>P1: short-stride jog and isometric exercise<br>P2: incorporated concentric and eccentric strengthening exercise, sprints<br>P3: eccentric strengthening with a power component, sprints | PATS:<br>P1: PATS exercises in frontal and transverse plane and icing<br>P2: PATS exercises in transverse and sagittal plane and icing<br>P3: PATS exercises in transverse and sagittal plane with increase speed and resistance and icing | 12 | TTRTP,The CC length of injury,Depth of the injured area, Strength, ROM, Pain |
| Askling et al.[6] | 75 | L-protocol 37<br>C-protocol 38 | L-protocol 25 ± 5<br>C-protocol 25 ± 6 | 69/6 | elite football | elite Swedish football players | 5 | physical examination and MRI | C-protocol: conventional exercises for the hamstrings with less emphasis on lengthening | L-protocol: loading the hamstrings during extensive lengthening, mainly during eccentric muscle actions | 12 | TTRTP,Re-injury |
| Askling et al.[5] | 56 | L-protocol 28<br>C-protocol 28 | L-protocol 21 ± 4<br>C-protocol 19 ± 3 | 38/18 | elite sprinters and jumpers | elite sprinters and jumpers | 5 | physical examination and MRI | C-protocol: conventional exercises for the hamstrings with less emphasis on lengthening | L-protocol: loading the hamstrings during extensive lengthening, mainly during eccentric muscle actions | 12 | TTRTP,Re-injury |
| Hagag et al.[20] | 35 | A 10<br>B 15<br>C 10 | A: 27.20 ± 3.71<br>B: 27.10 ± 3.70<br>C: 26.60 ± 4.25 | 35/0 | all types | athletes with 2nd degree tear | at one level | clinical assessment and sonography | - | A: PRP<br>B: LLLT<br>C: PRP + LLLT | - | Muscle healing,Angle of peak torque Function |

*(continued)*

Table 3. (continued)

| Study | N | allocation | Age (years) | Sex (M:FM) | Sports | Participants | Acute define (days) | Confirmed by | Relative control group | Intervention group | Follow-up (month) | Outcomes |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Sefiddashti et al.[50] | 37 | CT 19<br>CS 18 | CT 24.7 ± 4.1<br>CS 24.7 ± 3.9 | 21/16 | all types | athletes with grade 1 or 2 strain | 2 | resisted prone knee flexion and passive SLR test | CT: Ice at painful point | CS: Ice at painful point and hamstring static stretching | - | Pain, ROM, Functional status |
| Hickey et al.[24] | 43 | PF 22<br>PT 21 | PF 27.4 ± 5.2<br>PT 24.9 ± 5.3 | 43/0 | all types | athletes with a suspected HSI | 7 | clinical assessment | PF: hamstring strengthening exercise and progressive running with pain-free | PT: hamstring strengthening exercise and progressive running with pain-threshold limit | 6 | TTRTP,Re-injury rate, Flexibility, Strength, Fear of movement |
| Medeiros et al.[38] | 22 | LLLT 11<br>Placebo 11 | LLLT 30. 36 ± 7.06<br>Placebo 28.00 ± 7.42 | 22/0 | all types | athletes with grade1 and 2 injury | 5 | clinical assessment | Placebo: hamstring strength, trunk stabilization and movement agility | LLLT: hamstring strength, trunk stabilization and movement agility with low-level laser therapy | 6 | TTRTP,Number of sessions, Flexibility, Strength,Re-injury |

TTRTP: time to return to play; ROM: range of motion; STST: stretching and strengthening ; PAST: progressive agility and trunk stabilization; PRES: progressive running and eccentric strengthening; PATS: progressive agility and trunk stabilization; HI; hamstring injury; MRI: magnetic resonance imaging; CC: craniocaudal; L-protocol: lengthening protocol: C-protocol: conventional protocol; PRP: plasma rich platelets; LLLT: low level laser therapy; CT: cryotherapy; CS: cryostretching; SLR: straight leg raise; PF: pain-free; PT: pain-threshold; HSI: hastring strain injury.

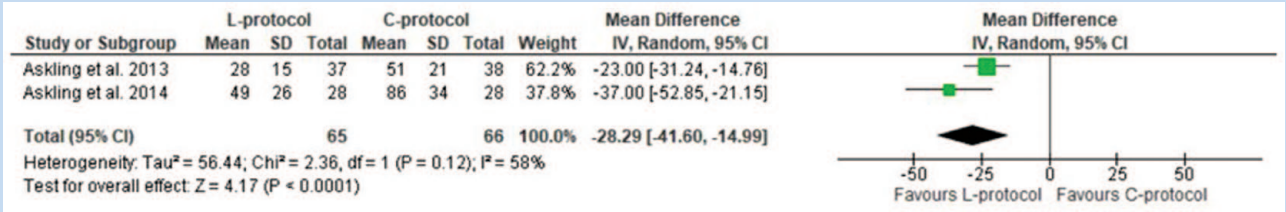


Figure 2. Forest plot of TTRTP between L-protocol and C-protocol. C, conventional; IV, inverse variance; L, lengthening; TTRTP, time to return to play.

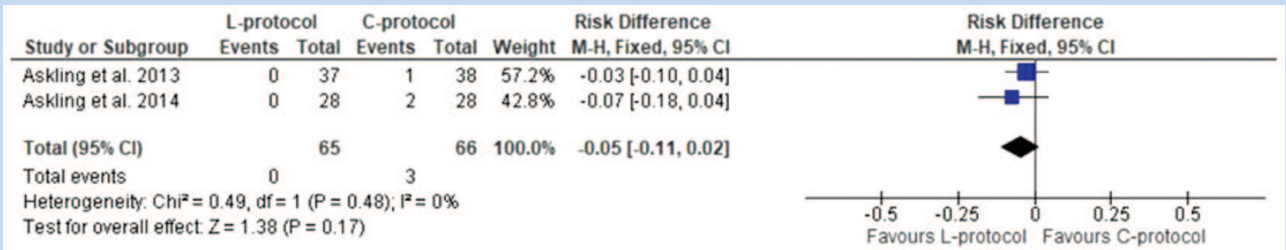


Figure 3. Forest plot of reinjury rate between L-protocol and C-protocol. C, conventional; L, lengthening; M-H, Mantel-Haenszel.

progressive sprint training in phase 3. Progressive resistance was added in trunk stabilization and the lunge walk as well as increased speed or/and resistance in phase 3 of the modified PATS program,[51] with frontal and transverse plane training in phase 1 and transverse and sagittal plane training in phase 2. Participants in all phases received 20 minutes of ice treatment after completing each exercise. The mean pain value during palpation at the starting point, 8.3 ± 3.0 and 9.9 ± 5.2 in the PRES and PATS groups, respectively, were reported to have dropped to 0.00 in both groups after completing the programs. The return-to-sport time was reported to be 28.8 ± 11.4 and 25.2 ± 6.3 days in the PRES and PATS groups, respectively, with no statistically significant differences ($P = 0.35$). However, the PATS exercise led to decreased reinjuries after a 1-year follow-up (5/13 in PRES and 2/16 in PATS), but the $P$ value was not shown.

Hickey et al[24] evaluated hamstring strengthening and progressive running protocols with therapists in patients at different pain levels twice a week until the participants met the predetermined RTP criteria. Participants in both groups performed the same exercise protocol. The only difference was the NRS of pain in the pain-free group and the pain-threshold group (≤4/10). The strengthening program, consisting of a hamstring bridge, a 45° hip extension, a Nordic hamstring exercise, and an eccentric slider, which was progressed with bilateral to unilateral exercise, mainly aimed at the biceps femoris long head fascicle length and eccentric knee flexor strength adaptation. Progressive running exercise followed the Silder[53] protocol and was progressed by varying the intensity (the hold distance and the acceleration and deceleration distances over a 50 m distance). The author reported that the median number of days until return to sports was 17 days in the pain-threshold group and 15 days in the pain-free group, which was not statistically significantly different ($P = 0.37$). An equal amount of reinjuries at a 6-month follow-up was reported: 2/22 in the pain-free group and 2/21 in the pain-threshold group, which was not significantly different ($P = 1.0$), but the pain-free group (day 23 and 27 after RTP) was reported to experience reinjuries later than the pain-threshold group (day 6 and 11 after RTP).

Medeiros et al[38] compared exercise-based rehabilitation programs following previous study methods with and without LLLT 3 times per week at a clinic.[5,13,51,53] The exercise program, which was performed similarly and progressed individually in both groups, was aimed toward enhancing hamstring strength, trunk stabilization, and movement agility. The participants in the LLLT group received continuous mode (7.5 $cm^2$ probe site), 5 diodes, 850 nm wavelength, peak power 100 mW, energy density 206.9 $J/m^2$, 1 minute per site on 3 sites on the hamstring muscles (at sites above and below the peak pain) immediately after each exercise session. The results showed that there was no statistically significant difference in time to return to sport ($P = 0.88$), 23.82 ± 12.62 in the pure exercise group and 23.09 ± 9.08 days in the exercise with LLLT group. At the 6-month follow-up, the participants in both groups demonstrated no sustained reinjuries.

Four studies reported the TTRTP outcome, but we cannot include the study of Hickey et al[24] due to the different value

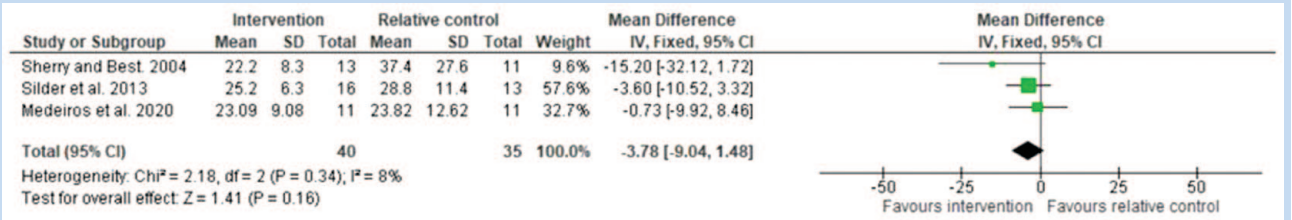


Figure 4. Forest plot of TTRTP between the intervention group and the relative control group. IV, inverse variance; TTRTP, time to return to play.

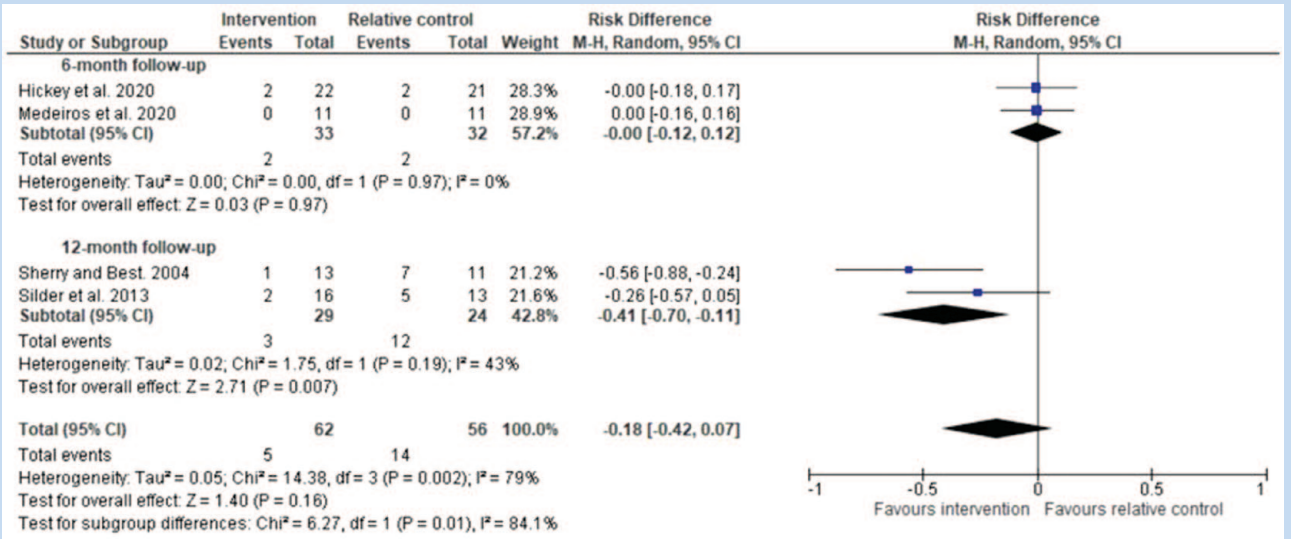


Figure 5. Forest plot of reinjury rate between the intervention group and the relative control group. C, conventional; L, lengthening; M-H, Mantel-Haenszel.

reported (median value). As such, only 3 studies can be pooled into meta-analysis. The mean difference with fixed effects model was used to estimate the number of days before the return to sport (or after injury). The pooled result estimates demonstrated intervention group (exercise and physical modalities) decreased the number of days to return to sport with mean difference -3.78 ([95% CI -9.04 to 1.48], $Z = 1.41$, $P = 0.16$). Statistical heterogeneity was not present ($I^2 = 8\%$) (Figure 4).

Four studies were pooled into the meta-analysis of reinjury rate for 2 different follow-up time points (at 6 and 12 months' follow-up). The pooled data at the 6-month follow-up showed that hamstring strengthening combined with a trunk stabilization exercise program and LLLT cannot prevent reinjury with an RD of -0.00 ([95% CI -0.12 to 0.12], $Z = 0.03$, $P = 0.97$). No statistical heterogeneity was found ($I^2 = 0\%$). The pooled treatment estimates from 2 studies in this domain evaluated at a 12-month follow-up after treatment highlighted significant effects in terms of mitigating reinjury with an RD of -0.41 ([95% CI -0.70 to -0.11], $Z = 2.71$, $P < 0.007$). Statistical heterogeneity was not present ($I^2 = 43\%$).

For the test of subgroup differences, there was a statistically significant difference in terms of a reinjury rate subgroup effect between the intervention group and the relative control group at $P = 0.01$ ([95% CI -0.42 to 0.07], $Z = 1.40$), meaning that different follow-up time points significantly modified the effect of the intervention in comparison with the control. Although there were small number of studies, and the participants contributed data to a 6-month follow-up subgroup and a 12-month follow-up subgroup, the results suggested that even with a limited number of included studies, it may still be possible to detect subgroup differences (Figure 5).

## Physical Modalities

Only 1 study with PEDro Scale scores demonstrating fair-quality applied pure modalities to treat acute hamstring injuries. Hagag et al[20] did not find a significant difference in pain with PRPs and LLLT for 3 groups: a single PRP injection, a single LLLT treatment, and a combination of these 2 treatments. The 2 groups treated with laser were applied 60 seconds/point beam radiation at the injury site with a wavelength 905 nm, a power

of 25 mW, and a dose of 1 J/cm$^2$ 3 times a week for 2 weeks. The authors reported that the single LLLT group or the laser combined with the PRP injection group significantly reduced pain in the acute hamstring injury, where the *P* value was <0.005 in both groups. Interestingly, there was no significant difference between the single LLLT and the laser combined with PRP injection groups ($P = 0.41$).

## DISCUSSION

### Main Findings

The meta-analysis of interventions for athletes with acute hamstring injuries revealed 9 RCTs, which included 396 participants where the PEDro quality assessment scores ranged from 3 to 9 (with 1 article scoring excellent, 2 articles good, 5 articles fair, and 1 article poor). In studies with results ranging from poor to fair quality, we found that there were no concealed allocations or blinding methods used. All of these studies included all types of sports, with the exception of the studies of Askling[5,6] studies, which evaluated the protocol in specific types of sports (football, sprinters, and jumpers). Different types of sports may affect the results due to differences in the injury mechanism. For example, high kicking is the injury mechanism for football players and ballet dancers, whereas stretching injuries are present in sprinters.[4] This has a direct impact on individual treatment methods as well as creating differences in the criteria for RTP clearance.

The designed exercise frequency in this meta-analysis varied across the studies. We found that the number of training days per week ranged between daily training to 5 days per week. Even if eccentric exercise had been confirmed to be harmless in the acute phase,[21] it should be realized that using eccentric muscle contraction in the inflammation phase may possibly exacerbate excessive muscle damage or delay recovery if the programs involve inappropriate management or improper workload progression.[41] Hence, the optimal eccentric exercise frequency and intensity should be specifically agreed upon in this type of injury so as to provide the most appropriate concept or framework to be applied in clinical settings.

Based on the results in the program reviews, a hamstring loading program can reduce duration of injury, whereas a program focused on strengthening using trunk stabilization and agility can prevent injury recurrence at a 12-month follow-up. The stretching program and solely physical modalities showed positive results but could not be pooled into the meta-analysis.

### Stretching Exercise With Physical Modalities

The efficacy of a stretching program was investigated in a prevention and rehabilitation program context.[28] It was found to have a superior effect when added to other programs rather than solely including conventional exercise or employing only a single modality. In 1 study, static stretching exercises were found to reduce TTRTP but did not lower reinjury rates.[37] Because this study, from the perspective of the PEDro scale, had a low quality score (score = 3), many readers may question the results after considering the study procedures, such as the allocation method, blinding method, or even the comparison method, since the results may be interpreted incorrectly. In contrast, a study identified as good quality (score = 7),[50] which used the same stretching protocol, had a superior effect on knee ROM and function but the procedure did not reduce pain during the rehabilitation period. From this study, we observed that the stretching dose (frequency and intensity) was adequate, but the period of time appeared to be too short (only 5 consecutive days) to see an effect on pain reduction during the acute phase. The acute phase thus could be extended to longer than usual (typically range 4-6 days)[2]. Adding more days to exercise or combining various exercise programs could possibly have led to better results in this study. We suggest that the effects of different types of stretching should be compared in hamstring injuries.

### Loading Hamstring Exercise During Extensive Lengthening

When comparing the current findings with a previous study conducted by Pas et al,[43] which computed the hazard ratio for the TTRTP ($P < 0.00001$) and used the risk ratio as the measure of reinjury ($P = 0.21$), we found that the results of the meta-analysis were similar, with a significant difference in only the TTRTP, where even different statistical analyses were used. The strength of these 2 studies was that the comparison of the L-protocol and C-protocol was performed specifically in track and field sports. The L-protocol demonstrated its superiority over the conventional program in terms of the TTRTP outcome and this may result from eccentric contraction of the L-protocol. Loading exercises in eccentric contraction may be rigorous and intensive in the acute phase. This might facilitate inflammation growth factors, delay recovery, and increase the risks of reinjury.[7,34] The exercise prescriptions involving intensity and volume in these 2 studies can be generalized directly to specific sport activities. However, the frequency, number of days of eccentric exercise, the intensity, and the dosage of exercise programs should be considered for each participant. It seems that the TTRTP in these 2 studies was longer than in the other studies in this meta-analysis. Nevertheless, the TTRTP results illustrated in the normal range of recovery time for football players (between 17.2 and 18.8 days) and runners (between 6 and 50 weeks).[3,14]

### Hamstring Strengthening Program Combined With Trunk Stabilization and Movement Agility With or Without Modalities

In this research, more studies were pooled, which made it possible to summarize the results of this kind of exercise.[43,47] The meta-analysis indicated that hamstring strengthening with trunk stabilization and movement agility after adding LLLT did not significantly decrease the duration of the injury, but the program can prevent a recurrence at 12-month follow-up. Only Silder et al[53] observed that this program can mitigate pain during palpation from the first day of training to after the full

rehabilitation program; however, statistical evidence was not presented. Previous fair-quality studies conducted by Sherry and Best[51] and Silder et al[53] showed that an intervention cannot significantly reduce TTRTP but can help prevent injury recurrence at 1-year follow-up. Also, the latest evidence from excellent and good-quality studies from Hickey et al[24] and Medeiros et al,[38] respectively, also did not show statistical significance for either TTRTP or the reinjury rate. Therefore, the program did not reduce TTRTP but prevented recurrence at a 12-month follow-up. Some limitations should be considered in this exercise domain. First, we could not pool the results of Hickey et al[24] into the meta-analysis in terms of the TTRTP outcome. Similarly, this study did not contain any new results that other studies had not already revealed and, as such, adding this study to our statistical analysis was unlikely to change our results. Second, the sample size calculation in the studies by Sherry and Best[51] and Silder et al[53] should be concerned because the relatively small sample size may not be enough to see the treatment effect. Finally, the frequency and time varied in the different studies, such as 5 days per week at home and 2 days per week at rehabilitation clinics.[24,53] This factor may have resulted in variations in the outcome measures in terms of exercise adherence and enhancement.

### Physical Modalities

In 1 study, only laser and cold therapy were applied to acute hamstring injuries. These 2 modalities are widely used and have been recommended to reduce pain due to soft tissue injuries in clinical settings.[2] Cold therapy is commonly used in standard clinical practice and is generally administered concurrently with therapeutic exercise programs to reduce inflammation resulting from injuries.[12] It is recommended that cryotherapy is applied for 20 minutes in the acute stage.[35] All studies that included cryotherapy in the programs mentioned 20 minutes of cold therapy after the exercise session. Laser therapy was found to be effective in the inflammatory phase.[10] LLLT was found to be equally effective for pain when compared to NSAID treatments in an animal study.[1] This meta-analysis showed the same results in reducing pain, when examining the effectiveness of LLLT compared with PRP injections. Lasers have shown enormous potential in soft tissue injuries; however, the application dosage may vary, and ensuring adequate doses is pivotal. Bjordal et al[10] recommended a clinical dose as follows: 7.5 J/cm$^2$ as the median value (range 0.3-19 J/cm$^2$), 5 to 171 mW/cm$^2$ as a power density with 632 to 660 nm wavelengths for continuous red lasers and 810 to 830 nm wavelengths for infrared lasers for acute pain. Two studies discussed in this paper, as aforementioned in the program reviews, used different doses but showed positive results in terms of reducing pain and facilitating return to sport.[20,38] The optimal doses of LLLT could not be concluded in this study due to the limited evidence. Laser doses should be investigated precisely for management of acute pain in hamstring injuries. Lastly, the other types of therapeutic modalities, such as neuromuscular electrical stimulation, ultrasound therapy, short wave diathermy, and shockwave therapy require more research and evidential support on their clinically curative effects in the first phase of such injuries.

### Limitations

There are several limitations in this study. First, we included only the very sound RCT study designs and limited the keywords, so very few studies were retrieved. Second, most studies included only male athletes, as high as 93.69% of participants. These results may thus not be generalizable to female athletes, because there is a difference in the knee flexor and extensor muscle morphology and injury mechanisms related to gender.[9,40] Third, as there were no actual control groups in the studies, we changed the keyword "control group" to "relative control group" in the strength training with trunk stabilization and agility domain to compare the treatment effects. However, we found that the relative control groups from this domain used the same protocol concepts but differed in some details, such as the number of phases and the number of repetitions during training. Only LLLT and cryotherapy were applied to the exercise programs, so we could not conclude which type of physical agents are appropriate for use in acute hamstring injuries. Also, we could not summarize the treatment dose for the same reason.

## CONCLUSION

The study aimed to investigate the effects of rehabilitation programs combined with modalities for treatment and prevention of injury recurrence in the acute phase of hamstring injuries. The synthesized evidence indicated that loading exercise is recommended for managing acute hamstring injuries. Hamstring strengthening with trunk stabilization and agility exercise prevented reinjury in a 1-year follow-up, but this was not clearly connected to TTRTP. Pain outcomes could not be concluded in this study. Evidence-based results regarding the types and dosages of modalities are insufficient and controversial in terms of managing acute hamstring injuries. This study demonstrated that exercise programs combined with physical modalities are as yet understudied in terms of examining their impact on pain, TTRTP, and reinjury rates.

## PERSPECTIVE

This is an updated review of the literature and a meta-analysis of the effectiveness of therapeutic exercise combined with physical modalities. Our findings confirmed that specific exercises for the hamstring management combined with therapeutic agents facilitate time to return-to-sports activities and prevent reinjuries. This study provides a robust model and framework for rehabilitation in the team management context. However, the standard protocols were inconclusive in this study. The exercise prescription and the use of therapeutic modalities treatments can vary based on a patient's condition. In addition, exercises should not only be prescribed after hamstring strain injuries but also be advocated to prevent reinjury.

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